

Normal Physiology – Question Bank Answer Key

Note: your sheet had question stems only (no answer options/key), so these are the standard textbook (Guyton/Ganong-level) answers with a one-line rationale each, numbered to match the original sheet.

1. **Total body Na^+ content** — Na^+ is the main osmotically active ECF solute; ECF volume tracks total body sodium.
2. **Homeostasis** — maintenance of a stable internal environment despite external change, via feedback control.
3. **Nucleus tractus solitarius (NTS)** — receives vagal afferents from gastric stretch receptors.
4. **The cell** — smallest structural/functional unit of life.
5. **Mitochondria** — site of oxidative phosphorylation/ATP synthesis.
6. **Gap junction – connexin** (tight junction – claudin/occludin; desmosome – cadherin/desmoplakin).
7. **Sodium (Na^+)** — principal extracellular cation.

8. **Intracellular fluid (ICF)** — ~2/3 of total body water.
9. **ICF is K⁺/protein-rich; ECF is Na⁺/Cl⁻-rich** — maintained by Na⁺/K⁺-ATPase.
10. **Na⁺/K⁺-ATPase** — primary active transporter (3 Na⁺ out, 2 K⁺ in per ATP).
11. **Simple diffusion** — not carrier-mediated, so no saturation kinetics.
12. **Na⁺/K⁺-ATPase pumps 3 Na⁺ out, 2 K⁺ in per ATP** — sets up the ionic gradients/RMP.
13. **Rapid Na⁺ influx** via voltage-gated Na⁺ channels — drives depolarization.
14. **K⁺ efflux** via voltage-gated (delayed rectifier) K⁺ channels — causes repolarization.
15. **Na⁺ channel inactivation** ($\pm$ persistent K⁺ conductance) — underlies the refractory period.
16. **H-band and I-band shorten** (A-band length unchanged) during contraction.
17. **The Na⁺-channel-inactivated phase** (depolarization spike through most of repolarization) = absolute refractory period.
18. **Failure of voltage-gated K⁺ (delayed rectifier) channels** to open.
19. **Center of the A-band**, where thin filaments overlap during full contraction.
20. **Binding of a new ATP molecule** to the myosin head.

21. **Troponin** – present in skeletal/cardiac muscle, absent in smooth muscle (which uses calmodulin/caldesmon).
22. **Pi release from myosin** causes the head to pivot, pulling actin toward the sarcomere center (power stroke).
23. **T-tubule depolarization is sensed by DHP receptors, mechanically coupled to RyR1** to release SR Ca^{2+} (no need for extracellular Ca^{2+} entry).
24. **SERCA pumps Ca^{2+} back into the SR**, lowering cytosolic Ca^{2+} , ending contraction.
25. **Troponin C** – binds Ca^{2+} to initiate cross-bridging.
26. **Cardiac muscle needs extracellular Ca^{2+} influx (CICR) to trigger SR release; skeletal muscle doesn't.**
27. **ACh binds nicotinic receptors on the motor end plate**, generating an end-plate potential.
28. **Gamma (γ) motor neurons** – innervate intrafusal fibers.
29. **GABA** (also glycine) – main inhibitory CNS neurotransmitter.
30. **Directly proportional** – thicker, myelinated fibers conduct faster.
31. **The neuron** – functional unit of the nervous system.

- 32. **Glutamate** – main excitatory CNS neurotransmitter.
- 33. **Neural (autonomic) and endocrine (hypophyseal) control**, integrating feedback signals.
- 34. **Cerebellum** – coordinates voluntary movement and posture.
- 35. **Initiating/planning movement and suppressing unwanted movement** – basal ganglia.
- 36. **Emotion, memory, and motivation** – limbic system.
- 37. **Chemical synapses** – transmission is one-way (pre- to postsynaptic).
- 38. **Acetylcholine** – from all preganglionic autonomic fibers (nicotinic receptors).
- 39. **Norepinephrine** – from postganglionic sympathetic fibers (except sweat glands = ACh).
- 40. **$A\alpha > A\beta > A\gamma > A\delta > B > C$** – descending conduction velocity.
- 41. **Caudate nucleus** (also putamen, globus pallidus, substantia nigra, subthalamic nucleus).
- 42. **Sympathetic fibers diverge widely and also trigger adrenal medullary catecholamine release**, which is cleared slowly from blood.
- 43. **β_2 -adrenergic receptors** – bronchial smooth muscle relaxation.
- 44. **Adrenal medulla, sweat glands, most blood vessels, piloerectors** – sympathetic-only

innervation.

- 45. **Contracts detrusor, relaxes internal sphincter** — promotes micturition.
- 46. **Miosis (pupil constriction)** is NOT sympathetic — it's parasympathetic; sympathetic causes mydriasis.
- 47. **Passive electrotonic (decremental) conduction** with summation at the soma.
- 48. **Receptor → afferent neuron → CNS integration center → efferent neuron → effector.**
- 49. **Monosynaptic reflex** — stretch/myotatic reflex.
- 50. **Muscle spindle (Ia afferents)** — detects length and rate of stretch.
- 51. **Golgi tendon organ** — senses muscle tension, provides protective autogenic inhibition via Ib afferents.
- 52. **Ca²⁺-calmodulin activates MLCK**, which phosphorylates myosin light chains to permit cross-bridging.
- 53. **Exaggerated extensor (decerebrate) rigidity** — loss of inhibitory descending input.
- 54. **Ib afferents from Golgi tendon organs** — mediate the inverse stretch reflex.
- 55. **Submandibular and otic ganglia** — parasympathetic supply to salivary glands.
- 56. **Iris** — colored part of the eye.

- 57. **Pacinian corpuscle** — rapidly-adapting (phasic) receptor.
- 58. **Olfactory epithelium (nose)** — detects odors.
- 59. **Eustachian tube** — drains middle/inner-ear-related fluid to the nasopharynx.
- 60. **Labeled-line principle** — modality is coded by which dedicated pathway/receptor fires, not the signal itself.
- 61. **Receptor/channel desensitization** — declining generator potential despite constant stimulus.
- 62. **Increased firing frequency (rate coding) and recruitment**, not AP amplitude.
- 63. **Slowly-adapting (tonic) receptors** — e.g., nociceptors, muscle spindles, baroreceptors.
- 64. **Light passes through ganglion and bipolar cell layers first**, reaching photoreceptors last (inverted retina).
- 65. **Superior temporal gyrus (Heschl's gyrus)** — auditory cortex.
- 66. **K⁺ influx** into hair cells (endolymph is K⁺-rich) — initiates depolarization.
- 67. **Left homonymous hemianopia** — right optic tract lesion.
- 68. **Place theory / tonotopic organization** of the basilar membrane.
- 69. **Difficulty localizing sound / processing tonal-musical information** (hearing loss is usually

mild/bilateral cortical input).

- 70. **Ciliary muscle relaxes, lens flattens** – reduced accommodation for distance viewing.
- 71. **Reduced rhodopsin (bleaching) and reduced cGMP-channel sensitivity** – light adaptation.
- 72. **Contralateral superior homonymous quadrantanopia** ("pie in the sky") – Meyer's loop lesion.
- 73. **Hemoglobin (in RBCs)** – main O₂ carrier.
- 74. **Systemic (non-renal, non-testicular) capillary beds** lack a countercurrent mechanism (unlike vasa recta/testis).
- 75. **Fibrinogen (and other clotting factors)** – present in plasma, absent in serum.
- 76. **Anemia / CO poisoning / decreased Hb** – reduce O₂-binding capacity.
- 77. **HbF (α₂γ₂)** – predominant fetal hemoglobin.
- 78. **HbF** – highest O₂ affinity (binds 2,3-BPG poorly).
- 79. **Decreased 2,3-BPG (↑Hb-O₂ affinity) and increased extracellular K⁺** – changes in stored blood.
- 80. **Electrolytes, albumin, glucose** – present in both serum and plasma (fibrinogen/clotting factors are plasma-only).
- 81. **A child with type O is genetically possible from A(het)×B(het) parents** – not evidence against parentage.

- 82. **HbF** — lowest 2,3-BPG affinity.
- 83. **↓Temperature, ↓2,3-BPG, ↓CO₂/alkalosis, HbF/CO presence** — shift curve left.
- 84. **Higher CO₂/H⁺ in venous blood (Bohr effect)**
lowers Hb-O₂ affinity.
- 85. **Local vasoconstriction (vasospasm)** — initial response to vessel injury.
- 86. **von Willebrand factor bridging platelet GPIb to exposed collagen.**
- 87. **GPIIb/IIIa receptor** (binds fibrinogen) — platelet-platelet aggregation.
- 88. **Factor VII has the shortest half-life** of the vitamin K-dependent factors, so PT (extrinsic) prolongs first.
- 89. **Platelet activation** — shape change, granule release, and aggregation (positive feedback).
- 90. **Protein C/Protein S** (inactivates Va, VIIIa) — also antithrombin III and TFPI.
- 91. **Biconcave, anucleate discs** — erythrocyte morphology.
- 92. **Carbohydrate (glycoprotein/glycolipid) antigens** on the RBC surface, set by glycosyltransferase genes.
- 93. **Lacks A, B, and Rh(D) antigens** — won't trigger ABO/Rh reactions in any recipient.
- 94. **Lewis antigens** (plasma-derived, adsorbed onto RBCs) can show altered expression with abnormal

plasma lipoproteins in liver disease.

- 95. **Anti-A/anti-B are pre-formed IgM antibodies present without prior exposure**, causing immediate hemolysis; Rh needs prior sensitization.
- 96. **Homozygous loss-of-function mutation in the H (FUT1) gene** — no H antigen, so A/B can't be expressed.
- 97. **AV node** — acts as the heart's "gate," delaying/filtering impulses from atria to ventricles.
- 98. **Arterioles** — greatest drop in systemic arterial BP.
- 99. **Ca²⁺ influx (L-type channels)** — maintains the cardiac plateau (Phase 2).
- 100. **Isovolumetric contraction** (also true of isovolumetric relaxation) — both valve sets closed, volume constant.
- 101. **Stimulates gastric HCl/pepsinogen secretion and gastrin release** — vagal cephalic phase.
- 102. **Loss of vagally-mediated receptive relaxation** — impairs liquid emptying after vagotomy.
- 103. **Hypotonic, low-volume, mucous-rich secretion at rest.**
- 104. **Peristaltic ("myenteric") reflex** — contraction proximal, relaxation distal to a bolus.
- 105. **Ductal cells reabsorb Na⁺/Cl⁻ (duct impermeable to water)** faster than they secrete K⁺/HCO₃⁻ —

dilutes saliva.

106. **Cholecystokinin (CCK)** – inhibits gastric emptying when fat enters duodenum.
107. **HCl (H⁺) secretion** is impaired – carbonic anhydrase supplies H⁺ in parietal cells.
108. **Secretin release** – stimulates pancreatic HCO₃⁻, inhibits gastric acid/motility.
109. **Gastric parietal cells** – source of intrinsic factor, required for B12 absorption in ileum.
110. **Cholecystokinin (CCK)** – stimulated by duodenal amino acids/peptides.
111. **Na⁺-glucose cotransport (SGLT1) apical, GLUT2 basolateral.**
112. **Highly acidic (pH ~1–2); contains HCl, pepsinogen, intrinsic factor, mucus.**
113. **Activates trypsinogen → trypsin**, which then activates other pancreatic proenzymes.
114. **Triglyceride hydrolysis** is impaired (lipase needed before micelle formation/absorption).
115. **GLUT5** – apical fructose transporter (facilitated diffusion).
116. **Emulsify fat and form micelles** to solubilize lipids/fat-soluble vitamins for absorption.
117. **Trypsinogen** (also chymotrypsinogen, procarboxypeptidase, prolipase) – secreted inactive.

118. **Secretin** — stimulates pancreatic ductal HCO_3^- secretion.
119. **Detoxification/drug metabolism and plasma protein synthesis** — liver functions unrelated to digestion.
120. **Conjugated hyperbilirubinemia, pale stools, dark urine, steatorrhea** — common bile duct obstruction.
121. **Carbohydrates (RQ = 1.0)** — highest respiratory quotient.
122. **Basal metabolic processes (mainly Na^+/K^+ -ATPase activity across organs)** — major resting energy expenditure.
123. **Renal corpuscles/glomeruli** — located in the cortex, not medulla.
124. **Vasodilation/natriuresis** — NOT effects of Angiotensin II (it vasoconstricts, releases aldosterone, promotes thirst/ADH).
125. **~125 mL/min** — normal GFR.
126. **Distal tubule and collecting duct** — site of urine acidification (H^+ secretion by intercalated cells).
127. **Pancreatic (secretin-stimulated) bicarbonate** — neutralizes duodenal HCl.
128. **Thick ascending limb of the loop of Henle** — impermeable to water under all conditions.
129. **Inulin** — freely filtered, not reabsorbed/secreted; its clearance = GFR, same behavior as fully-

blocked glucose.

- 130. **~600–700 mL/min** — effective renal plasma flow.
- 131. **Efferent arteriolar constriction** — most effectively raises GFR.
- 132. **Increased renin secretion** — direct JGA response to low pressure/flow.
- 133. **Reduced GFR decreases clearance** of urea/creatinine, causing accumulation.
- 134. **Maintained relatively constant (80–180 mmHg MAP)** via myogenic + tubuloglomerular feedback, independent of nerves/hormones.
- 135. **Regulates blood pressure/volume** via Na^+ and water retention plus vasoconstriction.
- 136. **↓Renal perfusion pressure, ↓NaCl at macula densa, ↑sympathetic (β_1) activity** — stimulate renin.
- 137. **Aldosterone** — increases urinary K^+ excretion.
- 138. **Reduced GFR / mild renal impairment.**
- 139. **Volume of plasma completely cleared of a substance per unit time.**
- 140. **Kidneys reabsorb filtered HCO_3^- and excrete titratable acid/ NH_4^+** , acting over hours-to-days.
- 141. **The glomerulus** — damage causes proteinuria + hematuria.
- 142. **Urea, creatinine, some ammonia present; no glucose/protein/blood/ketones** — normal urine.

143. **Hypertension, hypokalemia, metabolic alkalosis, low renin** — Conn's syndrome.
144. **GFR maintained/slightly ↑, RPF ↓** — filtration fraction increases.
145. **Effective renal plasma flow (RPF)** — represented by such a substance's clearance (e.g., PAH).
146. **Baroreceptor (↓stretch), macula densa (↓NaCl), and β₁-sympathetic signals** trigger JG cell renin release.
147. **Non-specific, immediate defenses present from birth** (barriers, phagocytes, complement, NK cells) — innate immunity.
148. **Antigen specificity and immunological memory** — distinguishes adaptive from innate.
149. **T lymphocytes** — cell-mediated immunity.
150. **~60–70% of MVV** — dyspnea appears when ventilatory reserve (dyspneic index) falls to about this level.
151. **ATP-PCr (phosphagen) system** — 100m dash energetics.
152. **Antigen-specific lymphocytes with memory (B/T cells)** are adaptive, not innate.
153. e.g., "**Passive artificial immunity — antibody injection,**" "**Active natural immunity — natural infection.**"
154. **IgM** — first antibody of the primary response.

- 155. **Complement doesn't produce antibodies** — it opsonizes, lyses (MAC), and drives chemotaxis/inflammation.
- 156. **Defend against pathogens while distinguishing self from non-self.**
- 157. **Immunological memory.**
- 158. **Type III hypersensitivity (immune complex-mediated)** — serum sickness.
- 159. **Hypothalamus (preoptic/anterior region)** — thermoregulatory center.
- 160. **Vasoconstriction, shivering, ↑metabolic heat production, piloerection** — response to cold.
- 161. **Radiation** — main heat loss route at rest, normal room temperature.
- 162. **Heat-loss responses (vasodilation, sweating)** — anterior hypothalamus stimulation.
- 163. **Sweating and cutaneous vasodilation** — prevent hyperthermia in heat.
- 164. **Voluntary behavior** (seeking shade, removing clothes) — behavioral thermoregulation.
- 165. **↑Thyroid hormone (T3/T4)** — raises metabolic heat production long-term in cold.
- 166. **Local axon-reflex vasodilation/vasoconstriction** (e.g., cold-induced vasodilation, "hunting reaction") — local cutaneous thermal reflex.
- 167. **High humidity reduces evaporation**, so sweat drips off without cooling.

- 168. **Core temp >40°C with CNS dysfunction**, often with cessation of sweating – heat stroke.
- 169. **Shivering thermogenesis + vasoconstriction** (plus non-shivering thermogenesis in infants) – prevent hypothermia.
- 170. **Endogenous pyrogens (IL-1, IL-6, TNF) → hypothalamic PGE2 → raised set point** – first events of fever.
- 171. **Cerebellum (vestibulocerebellum) and vestibular nuclei** – posture/equilibrium.
- 172. **Initiates/plans movement, suppresses unwanted movement** – basal ganglia.
- 173. **Balance and eye-movement/postural control** – vestibulocerebellum.
- 174. **Muscle spindle stretch reflex (gamma loop)** – maintains muscle tone.
- 175. **Ataxia, intention tremor, dysmetria, hypotonia, nystagmus** – cerebellar damage.
- 176. **Vestibulospinal tract** – adjusts posture relative to gravity.
- 177. **Homeostasis** – stable internal environment concept.
- 178. **e.g., insulin-glucose regulation or thermoregulation** – negative feedback example.
- 179. **Nervous system** – fastest control system (vs. slower endocrine system).

- 180. **Touch, pressure, temperature, pain, proprioception carried via DCML and spinothalamic tracts.**
- 181. **Meissner's corpuscles** (and Merkel discs) – fine touch/texture.
- 182. **Anterolateral (spinothalamic) tract** – pain pathway.
- 183. **Substance P** (and glutamate) – spinal pain transmission.
- 184. **Prostaglandins, bradykinin, histamine, substance P** – inflammatory pain mediators.
- 185. **Sustained bilateral muscle contraction (neck/scalp), non-pulsatile pain** – tension headache.
- 186. **Hypoxia ($\downarrow\text{PaO}_2$), via carotid body chemoreceptors** – main stimulus at altitude.
- 187. **Hyperventilation, respiratory alkalosis, $\uparrow\text{HR}$, initial $\downarrow\text{plasma volume}$** – acute altitude changes.
- 188. **$\uparrow\text{EPO/RBC mass}$ (polycythemia), $\uparrow\text{2,3-BPG}$, $\uparrow\text{capillary density}$, $\uparrow\text{mitochondria}$** – chronic altitude adaptation.
- 189. **Lower lean muscle mass/higher fat %** – explains lower female BMR.
- 190. **$\uparrow\text{HR}$, $\uparrow\text{cardiac output}$, $\uparrow\text{systolic BP}$, $\uparrow\text{ventilation}$, **blood flow shift to muscle**** – moderate exercise.
- 191. **Suppresses appetite/increases energy expenditure** – leptin signals adequate fat stores.

192. **PaO₂ itself stays low (lower barometric pressure)**
even though $\uparrow$ Hb/RBC mass raises O₂-carrying capacity.
193. **$\uparrow$ Red cell mass (polycythemia)** – chronic altitude O₂ delivery adaptation.
194. **Residual volume and arterial stiffness/systolic BP** – increase with aging (most other parameters decline).
195. **Flow = Δ Pressure / Resistance** – hemodynamic "Ohm's law."
196. **Radius⁴ (r⁴)** – Poiseuille's law.
197. **16-fold increase** – resistance $\propto 1/r^4$, so halving r $\rightarrow 2^4 = 16\times$.
198. **Arterial baroreceptor reflex** – most important short-term BP control.
199. **$\uparrow$ Sympathetic outflow $\rightarrow$ tachycardia + vasoconstriction** – immediate response to a BP drop.
200. **$\downarrow$ Hematocrit (anemia) or $\downarrow$ plasma proteins** – lowers viscosity, raises flow.
201. **L-type Ca²⁺ channel inactivation + delayed rectifier K⁺ efflux** – ends the ventricular plateau.
202. **Extracellular Ca²⁺ influx (L-type channels) triggering CICR from the SR.**
203. **$\uparrow$ Preload (venous return/EDV) $\rightarrow$ $\uparrow$ stroke volume**
– Frank-Starling relationship.

- 204. **Funny current (If, HCN channels)** (plus T-type Ca^{2+}) – Phase 4 pacemaker depolarization.
- 205. **Ventricles relax, semilunar valves close, pressure falls rapidly before AV valves open – volume constant** – isovolumetric relaxation.
- 206. **Cardiac muscle requires extracellular Ca^{2+} entry (CICR) and is a functional syncytium (gap junctions); skeletal doesn't need extracellular Ca^{2+} .**
- 207. **Long absolute refractory period (from the prolonged plateau) prevents summation/tetany.**
- 208. **Sympathetic (β_1) stimulation and vagal withdrawal** – increase heart rate.
- 209. **Venous return (preload)**, via the Frank-Starling mechanism – key immediate determinant of CO/SV.
- 210. **$\uparrow$ Afterload or extreme $\uparrow$ HR ($\downarrow$ filling time)** – can decrease CO.
- 211. **Intrinsic length-tension property: $\uparrow$ stretch (preload) $\rightarrow$ $\uparrow$ contractile force**, matching CO to venous return.
- 212. **$\uparrow$ BP $\rightarrow$ $\uparrow$ baroreceptor firing $\rightarrow$ $\downarrow$ sympathetic/ $\uparrow$ parasympathetic $\rightarrow$ $\downarrow$ HR/BP** – fast, negative feedback, resets with chronic hypertension.
- 213. **CO regulated by preload, afterload, contractility, and HR**, matching venous return.

- 214. **↑HR and peripheral vasoconstriction** — baroreceptor response on standing.
- 215. **↑Sympathetic vasoconstriction (↑TPR)** — most effective short-term MAP raiser.
- 216. **↑Venous return (Frank-Starling) and circulating catecholamines** — CO can still rise in a denervated heart.
- 217. **Arterioles** — resistance vessels.
- 218. **Veins** — capacitance vessels.
- 219. **Vessel radius (r^4)** — greatest influence on flow.
- 220. **Capillaries** — highest total cross-sectional area, lowest velocity.
- 221. **Capillaries** — small radius keeps wall tension low (Laplace's law) despite thin walls.
- 222. **Arterioles** — biggest contributor to total peripheral resistance.
- 223. **Diaphragm** (with external intercostals) — quiet inspiration.
- 224. **Passive elastic recoil** of lungs/chest wall — normal quiet expiration.
- 225. **Intrapleural pressure** — always negative in quiet breathing.
- 226. **Arterial PCO_2** (via central chemoreceptors/CSF H^+) — main respiratory drive normally.
- 227. **Reduces alveolar surface tension; produced by type II pneumocytes; stabilizes alveoli** —

surfactant.

- 228. **Inversely proportional** — $\uparrow$ alveolar ventilation $\rightarrow$ $\downarrow$ PaCO₂.
- 229. **Thoracic expansion (diaphragm contraction)**
increasing lung volume/elastic recoil — more negative intrapleural pressure.
- 230. **Perfusion falls more than ventilation from base to apex (gravity)** — higher V/Q at apex.
- 231. **$\uparrow$ CO₂/H⁺ in tissue capillaries lowers Hb-O₂ affinity**, promoting O₂ unloading (Bohr effect).
- 232. **Residual volume** (and FRC, TLC, which depend on it) — not measurable by simple spirometry.
- 233. **Hypothermia/ $\downarrow$ 2,3-BPG/alkalosis** left-shift the curve — NOT rightward shifters.
- 234. **~75%** — Hb saturation at PO₂ ~40 mmHg (mixed venous).
- 235. **~98% of O₂ is bound to hemoglobin**, small fraction dissolved in plasma.
- 236. **Humidification and continuous gas exchange (O₂ uptake/CO₂ addition)** dilute alveolar PO₂ below inspired PO₂.
- 237. **V/Q mismatch, diffusion impairment, right-to-left shunt** — increase the A-a gradient.
- 238. **~250 mL/min** — resting VO₂.
- 239. **Bicarbonate (~70%)** — main CO₂ transport form.

240. **Ratio of O₂ consumption to O₂ delivery** – determines the O₂ extraction ratio.
241. **Medulla oblongata (pre-Bötzinger complex/dorsal respiratory group)** – respiratory rhythm generator.
242. **CO₂ (via H⁺ in CSF)** – main central chemoreceptor stimulus.
243. **Opioids** (also barbiturates/sedatives) – depress respiratory centers, cause arrest in overdose.
244. **Lung stretch receptors inhibit further inspiration when overinflated** – Hering-Breuer reflex.
245. **Peripheral (carotid body) chemoreceptor response to hypoxemia** – initial altitude ventilation increase.
246. **Rising PaCO₂/falling pH** (central chemoreceptor drive) – main trigger to resume breathing after breath-holding.
247. **Renal excretion of HCO₃⁻ normalizes CSF pH**, removing the brake so hypoxic drive keeps ventilation elevated.
248. **Promotes sustained inspiration** – apneustic center (lower pons).
249. **Lung inflation → stretch receptors → vagal inhibition of inspiration** – Hering-Breuer inflation reflex.
250. **Steroid and thyroid hormones** – act via intracellular receptors on gene transcription.

251. **IP₃/DAG–Ca²⁺ (phospholipase C pathway)** – second messenger for V1/α1-receptor signaling.
252. **Growth hormone** (and prolactin) – use JAK-STAT signaling.
253. **Oxytocin and ADH** – made in hypothalamus, released from posterior pituitary.
254. **TRH→TSH→T3/T4, with T3/T4 negative feedback** on TRH and TSH.
255. **↓Serum calcium by inhibiting osteoclasts** (minor role in adult humans) – calcitonin, from thyroid C-cells.
256. **Lipophilic hormone diffuses in, binds intracellular receptor, complex acts as a transcription factor.**
257. **Lactotrophs (prolactin)** – under predominantly inhibitory (dopamine) hypothalamic control.
258. **TSH, LH, FSH, hCG** – share a common α-subunit, unique β-subunit gives specificity.
259. **Central (neurogenic) diabetes insipidus.**
260. **↑Ca²⁺ reabsorption (distal tubule), ↓phosphate reabsorption, activates vitamin D** – PTH renal actions.
261. **Primary hyperparathyroidism.**
262. **ACTH** (from CRH-driven anterior pituitary) – stimulates cortisol.
263. **↑Gluconeogenesis/blood glucose, anti-inflammatory, protein catabolism** – cortisol effects.

264. **Angiotensin II** (and plasma K^+) – major aldosterone regulators.
265. **Glomerulosa–mineralocorticoids, fasciculata–glucocorticoids, reticularis–androgens.**
266. **High cortisol with LOW (suppressed) ACTH** – adrenal-tumor Cushing's.
267. **PNMT (phenylethanolamine N-methyltransferase)** – $NE \rightarrow Epi$, cortisol-induced.
268. **β -adrenergic receptors (β_2 liver glycogenolysis; β_1/β_3 adipocyte lipolysis).**
269. **Sympathetic preganglionic (cholinergic, nicotinic) stimulation** – triggers catecholamine release in stress.
270. **Hypertension, hypokalemia, metabolic alkalosis, low renin** – primary hyperaldosteronism.
271. **Episodic massive catecholamine release from the tumor** – pheochromocytoma's paroxysmal symptoms.
272. **Cholesterol $\rightarrow$ pregnenolone (cholesterol side-chain cleavage/desmolase, StAR-mediated)** – rate-limiting step.
273. **11β -hydroxylase deficiency** – DOC accumulation, hypertension, low cortisol.
274. **Ovulation** (completion of meiosis I, follicle rupture) – triggered by LH surge.
275. **Progesterone** (from corpus luteum) – maintains secretory endometrium.

276. **Day 14** – LH surge in a 28-day cycle.
277. **Oogonia arrest in prophase I before birth; meiosis I completes just before ovulation; meiosis II completes only at fertilization.**
278. **Corpus luteum** – main progesterone source, first trimester.
279. **Human placental lactogen (hPL).**
280. **Metaphase II** – secondary oocyte arrest.
281. **Sustained high estradiol switches hypothalamic-pituitary feedback from negative to positive, triggering the LH surge.**
282. **Falling progesterone (and estrogen)** – first hormonal change if corpus luteum fails.
283. **Progesterone/estrogen withdrawal → spiral artery constriction → endometrial ischemia/shedding.**
284. **Rising estrogen initially gives negative feedback, then switches to positive feedback near ovulation.**
285. **Human placental lactogen (hPL).**
286. **hCG secretion by trophoblast** – earliest endocrine change, maintains corpus luteum.
287. **Placenta relies on fetal/maternal precursors (DHEA-S) for estrogen synthesis – the "fetoplacental unit."**
288. **Progesterone** (with estrogen, prolactin, hPL) – mammary lobuloalveolar growth.

289. **Spermatogonia** – germ cells that undergo spermatogenesis.
290. **Testosterone** (with FSH) – critical for initiating/maintaining spermatogenesis.
291. **Sertoli cells** – FSH target supporting spermatogenesis.
292. **LH** – stimulates Leydig cells to make testosterone.
293. **Peripheral tissues (prostate, genital skin)**, via 5 α -reductase – testosterone→DHT.
294. **Inhibin** (from Sertoli cells) – inhibits FSH.
295. **Spermatogonia → primary spermatocyte → secondary spermatocyte → spermatid → spermatozoon.**
296. **Sertoli-cell tight junctions isolate germ cells from the immune system** – blood-testis barrier.
297. **Testosterone inhibits GnRH/LH; inhibin inhibits FSH** – negative feedback loop.
298. **Stimulates gastric HCl/pepsinogen and gastrin release** – cephalic phase vagal effect (same as Q101).
299. **Loss of vagal receptive relaxation** – impaired liquid emptying post-vagotomy (same as Q102).
300. **Hypotonic, low-volume, mucous-rich** resting saliva (same as Q103).
301. **Peristaltic reflex** – contraction proximal, relaxation distal to a bolus (same as Q104).

302. **Non-specific, immediate, present from birth** – innate immunity (same as Q147).
303. **Antigen specificity and memory** – adaptive vs innate (same as Q148).
304. **T lymphocytes** – cell-mediated immunity (same as Q149).
305. **Secondary response is faster, larger, and IgG-predominant** (memory B cells) vs slower IgM-predominant primary response.
306. **Antigen-specific lymphocytes with memory** are adaptive, not innate (same as Q152).
307. **e.g., "Passive artificial immunity – antibody injection"** (same as Q153).
308. **IgM** – first antibody, primary response (same as Q154).
309. **Complement doesn't make antibodies** (same as Q155).
310. **Passive natural immunity** – maternal IgG crossing the placenta.
311. **Cell-mediated (T-cell) immunodeficiency** – e.g., DiGeorge syndrome, thymic aplasia.
312. **APCs process/present antigen via MHC to T cells**, initiating adaptive immunity.
313. **Type III hypersensitivity (immune complexes)** – serum sickness (same as Q158).
314. **Hypothalamus** – thermoregulatory center (same as Q159).

- 315. **Vasoconstriction, shivering, ↑heat production** — cold response (same as Q160).
- 316. **Radiation** — resting heat loss (same as Q161).
- 317. **Heat-loss responses (vasodilation, sweating)** — anterior hypothalamus (same as Q162).
- 318. **Sweating and cutaneous vasodilation** — heat response (same as Q163).
- 319. **Voluntary behavior** — behavioral thermoregulation (same as Q164).
- 320. **↑Thyroid hormone** — long-term cold response (same as Q165).
- 321. **Local axon-reflex vasomotor response** — cutaneous thermal reflex (same as Q166).
- 322. **High humidity blocks evaporative cooling** (same as Q167).
- 323. **Core temp >40°C with CNS dysfunction** — heat stroke (same as Q168).
- 324. **Shivering + vasoconstriction** — prevents hypothermia (same as Q169).
- 325. **Endogenous pyrogens → hypothalamic PGE2 → raised set point** — fever onset (same as Q170).
- 326. **Cerebellum/vestibular nuclei** — posture/equilibrium (same as Q171).
- 327. **Initiates/refines movement** — basal ganglia (same as Q172).

328. **Balance/eye-movement control** – vestibulocerebellum (same as Q173).
329. **Muscle spindle gamma loop** – muscle tone (same as Q174).
330. **Ataxia, intention tremor, dysmetria, hypotonia** – cerebellar damage (same as Q175).
331. **Vestibulospinal tract** – balance/posture (same as Q176).
332. **Homeostasis** (same as Q177).
333. **e.g., insulin-glucose feedback** (same as Q178).
334. **Nervous system** – fast control (same as Q179).
335. **Touch, pressure, temperature, pain, proprioception** – somatic sensations (same as Q180).
336. **Meissner's corpuscles** – fine touch/texture (same as Q181).
337. **Spinothalamic (anterolateral) tract** – pain (same as Q182).
338. **Substance P** – spinal pain neurotransmitter (same as Q183).
339. **Prostaglandins/bradykinin/histamine** – inflammatory pain mediators (same as Q184).
340. **Sustained muscle contraction, bilateral, non-pulsatile** – tension headache (same as Q185).
341. **Hypoxia via carotid bodies** – altitude ventilation stimulus (same as Q186).

342. **Hyperventilation, respiratory alkalosis, ↑HR** – acute altitude change (same as Q187).
343. **↑RBC mass, ↑2,3-BPG, ↑capillary density** – chronic altitude adaptation (same as Q188).
344. **Lower lean muscle mass** – lower female BMR (same as Q189).
345. **↑HR, CO, systolic BP, ventilation** – moderate exercise (same as Q190).
346. **Suppresses appetite** – leptin (same as Q191).
347. **PaO₂ itself remains low** despite ↑ventilation – altitude finding (same as Q192).
348. **↑Red cell mass** – O₂ delivery adaptation (same as Q193).
349. **Residual volume/arterial stiffness increase** with age (same as Q194).
350. **Flow = Δ Pressure/Resistance** (same as Q195).